SAMPLE

Cosmos Unveiled: Galaxies

Cities of Stars Across the Universe

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Introduction

The scale the reader is entering calls for one preliminary, offered as entertainment rather than warning: the subject of this book is the largest class of objects that can be studied individually. The stars are individuals, but they are parts; the planets, individuals, but small parts of parts. The galaxies are the largest things in nature that have names, that can be photographed whole, that have childhoods and collisions and biographies, and the reader about to meet them is entering the uppermost story of the physical world, the one above which only the universe-as-a-whole remains, and the universe-as-a-whole has no face to photograph, while the galaxies do, which is why the family album, in this book and in the science, is full of portraits.

On a clear night far from city lights, a pale band of light arches across the sky, and for most of human history that band was the whole of the subject. We now know it is an inside view: the band is our own galaxy seen edge-on from within, a vast star system of which our sun is one member among hundreds of billions. And we know something that took our species much longer to establish: the Milky Way is not alone. The count of naked-eye galaxies opens in an early capsule, the reader's own sky being the example: the unaided eye, under a dark sky, can see one galaxy beyond our own, the great Andromeda system, easily in autumn, plus the two southern satellites of the Milky Way for readers beneath the equator, so that the entire extragalactic universe visible to unaided human sight, throughout all history until a century ago, was a single object whose nature was unknown, and the telescope's census, from a few objects to a few hundred to billions, is the instrument's own biography, a fact the discovery chapter tells in full.

The small fuzzy patches that dot the dark sky, dismissible to the naked eye as faint smudges or mislabeled clouds, are other galaxies, each one a complete star system, an island of stars and gas and dark matter, separated from its neighbors by distances so vast that light itself is the only courier fast enough to carry news between them. The universe holds trillions of these islands, and their story is the subject of this book.

The definition's one refinement needs its early flag, because the book's honesty requires it: the boundary of "galaxy" is a convention at the edges. The largest globular clusters, tight balls of a million old stars orbiting a galaxy's halo, are in content miniature versions of the smallest dwarf galaxies, and the classification of borderline systems, the ones that are barely galaxies or barely clusters, is a standing taxonomic argument the field enjoys, with new discoveries in the reader's own neighborhood still arriving, some objects having been promoted or demoted within the last decade. The family's boundaries, like most families', are fuzzier than its center, and the reader should treat the book's clean categories as honest centers with honest edges rather than as walls.

A capsule image begins the catalog honestly: a galaxy is to the universe roughly what a cell is to a body, the unit of structure at the scale where the body's larger architecture begins, and the comparison survives the numbers, since the family's members, like a body's cells, come in types with different functions, live in different neighborhoods, and share a common origin in a single early template, all points the later chapters earn. The catalog that follows, the family's contents, sizes, numbers, and distances, is the unit's specification, the reader's datasheet for everything else the book contains.

A galaxy, stated plainly for a start, is a gravitationally bound system of stars, stellar remnants, gas, dust, and dark matter, ranging from systems of a few million stars to giants of a hundred trillion, spanning from a few thousand to hundreds of thousands of light-years. The word descends from the Greek name for our own band of light, itself tied to the myth of spilled milk across the heavens, and the modern usage generalizes the family of which the Milky Way is one member. Galaxies are the universe's fundamental building blocks, the units into which matter has organized on the intermediate scale, and every large thing the reader has ever heard of in astronomy, every star, every planet, every nebula, lives inside one.

The subject rewards a full telling because galaxies braid together so many threads of modern science. They are the homes of stars, and their gas supplies are the raw material from which every subsequent generation condenses. They are the tracers of the cosmic web, the lattice of large-scale structure that this series' cosmology volumes describe, so that mapping galaxies is mapping the universe itself. They are the arenas where dark matter shows its hand, the nurseries where the first stars lit, the laboratories where black holes grow to monstrous size, and the stage on which our own galaxy's biography, collision by collision, has been written. A book about galaxies is therefore a book about nearly everything, and the chapters ahead are arranged to make that breadth manageable: the nature and anatomy of galaxies first, then their discovery, their hidden mass, their formation and life cycles, their central engines and their collisions, and finally the modern research programs and open questions

that define the field today.

A note on scale-words, stated once for the whole book: the numbers here are large beyond intuition, and the text will occasionally pause to rebuild them in human terms, but the reader's safest habit is to keep a few anchors and let the rest be relative. A light-year is a distance, the span light crosses in a year, about ten trillion kilometers; the sun's light takes about eight minutes to reach Earth, the nearest other star's four years, the galaxy's center tens of thousands of years, and the nearest other galaxy's light longer than the whole human story. That one ladder, minutes to millions of years, is the subject's entire geography, and every number in the chapters ahead is a rung on it, so the reader who holds the ladder reads the rest of the book without vertigo.

One promise before the journey begins, made once and kept throughout: the numbers in this book are real measurements, the history is attributed to its real protagonists, and the genuinely debated questions, of which the field has several, are labeled as debates rather than smoothed into false certainty. Galaxies are old, vast, and strange, but they are also among the best-studied objects in nature, and the reader who finishes this book will know them the way modern astronomy knows them: as magnificent structures, measured with increasing precision, still keeping a few secrets.

Chapter 1: Understanding Galaxies: The Island Universes

The place to begin is with the objects themselves, cataloged honestly: what a galaxy contains, how large the family runs, and what the numbers are that describe them.

A galaxy's contents are worth listing, because the list is the chapter's spine. Stars, first: from millions in the smallest dwarfs to hundreds of billions in the giants, in every stage of life, birth in gas clouds to death as white dwarfs, neutron stars, and black holes. Planetary systems, second: the accumulated wreckage of star formation, disks of dust and gas around young stars, condensed into planets, asteroids, and comets; our own system is one of billions within our galaxy alone. Interstellar gas, third: the diffuse medium between the stars, thin and cold in some places, hot and turbulent in others, the raw material from which new stars form, and the medium through which old stars return their substance when they die. Dust, fourth: tiny solid grains, a small fraction of the mass but a large influence, dimming and reddening starlight and glowing in the infrared where the coldest clouds are studied. And dark matter, fifth and dominant: the invisible substance, outweighing all the rest together several times over, whose gravity holds each system bound and whose nature remains one of physics' great open problems.

The family's size range is best told in honest numbers. A capsule comparison for the giants completes the range: the largest ellipticals, the cluster-thrones described in the family album chapter, span a million light-years or more, ten times the Milky Way's disk, and hold a trillion suns, with a single such system outweighing the entire local group, and their central black holes, billions of suns, are the class's records too. The family's range, from the dwarfs' millions to the giants' trillions, is thus a factor of a million in stars, and the reader's own galaxy, mid-range in a family that spans six orders of magnitude, is the standard the book's numbers are calibrated on, a provincial comfort worth stating once.

The smallest recognized galaxies, the dwarf spheroidals and the ultra-diffuse systems the later chapters describe, hold mere millions of stars in bodies a few thousand light-years across, so faint that new ones are still being found in our own backyard. The giants, the great ellipticals at the centers of galaxy clusters, hold a thousand times more, spanning hundreds of thousands of light-years and shining with the combined light of a hundred billion suns. Our own galaxy, the Milky Way, sits near the middle of the range, a broad disk a hundred thousand light-years across containing on the order of a few hundred billion stars, and the reader should hold the calibration: "normal" covers a factor of a thousand in both mass and size, and our own system, though huge by human standards, is thoroughly typical of the cosmic population.

The numbers of galaxies are the subject's most vertiginous facts, and the modern counts are worth stating. The observable universe contains, by the deep-sky surveys' census, on the order of a hundred billion galaxies within reach of our largest telescopes, and the total, extrapolated beyond the surveys' reach, is plausibly some two trillion, a figure the field quotes with its caveats, since counting the smallest, dimmest systems is an ongoing struggle. The universe is thus, in a real sense, mostly galaxies: the matter that is not in the rarefied gas between them is overwhelmingly bound inside these systems, and the reader looking at the night sky is looking at a sample of one island out of the largest archipelago there is.

A capsule census of the local family completes the orientation, because the reader's own neighborhood is the

book's recurring example. The local group, the congregation of a few dozen galaxies to which the Milky Way belongs, spans a few million light-years and holds two large spirals, ourselves and the great Andromeda system, a mid-size spiral or two, and the rest an entourage of small dwarfs, the family's typical shape: a couple of capitals with a cloud of small attendants. The group's total mass is dominated by its two spirals and their halos, and the group's future, the mergers chapter describes, is a slow approach of the two capitals toward union, so the reader's own address is both typical and eventful, a modest congregation with a marriage scheduled, and the book will return to it as the standard example of everything the family does.

The idea is older than the evidence by more than a century, and that fact belongs in the orientation. In the middle 1700s, when the nebulae were only smudges in the small telescopes of the day, a philosopher reasoned from the Milky Way's own nature to the possibility that the diffuse patches were similar systems seen from outside, one speculative page in a cosmology otherwise built of guesses. The reasoning was sound but untestable: no instrument of the era could settle it, so the hypothesis waited through the whole of the nineteenth century while the smudges accumulated in the catalogs as objects of no settled kind. The subject's first lesson sits right there: right answers in this field arrive only with right instruments, and an idea can be correct, published, and entirely unusable for a hundred and fifty years.

The distances between the islands close the chapter, because they define the subject's character. Our nearest large neighbor lies about two and a half million light-years away, so far that its light, arriving tonight, left before the genus Homo existed on Earth, and the typical separations between galaxies in our neighborhood are a few times that, meaning the galaxies are small compared to the distances between them, like motes separated by rooms. A collision between two galaxies is therefore a slow, rare, and magnificent event rather than a crash, the mergers chapter describes it, and a galaxy's life is mostly a life alone, quietly converting its gas to stars while its neighbors drift at a distance. The islands are real islands, and the space between them is the sea, and the reader who holds that image holds the subject's essential loneliness, softened by the knowledge that the islands do, over immense timescales, meet.